

Total No. of Questions : 8]

SEAT No. :

P644

[Total No. of Pages : 3

[6004]-605

B.E. (Mechanical)

DYNAMICS OF MACHINERY

(2019 Pattern) (Semester - VII) (402042)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagram must be draw wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) A horizontal spring mass system with coulomb damping has a mass of 5 kg attached to a spring of stiffness 980 N/m. If the coefficient of friction is 0.25, calculate: **[8]**

- i) the frequency of free oscillations.
 - ii) the number of cycles corresponding to 50% reduction in amplitude if the initial amplitude is 5cm.
 - iii) time taken to achieve this 50% reduction.
- b) Explain with neat diagram mathematical model of a motor cycle. **[5]**
- c) By using energy method, find the natural frequency of undamped free longitudinal vibrations **[4]**

OR

Q2) a) Derive an expression for the motion of spring-mass dashpot system in case of: **[8]**

- i) over damped system
 - ii) critically damped system
 - iii) under damped system.
- b) A flywheel of mass 10 kg and radius of gyration 0.3 m makes torsional rotations under a torsion spring of stiffness 5 Nm/rad. A viscous damper is fitted and it is found that the amplitude is reduced by a factor 100 over any two successive cycles. Find **[5]**
- i) Damping factor
 - ii) Damping coefficient
 - iii) Damped frequency
 - iv) Periodic time oscillation
- c) A mass of 3kg is supported on an isolator having a spring constant of 3000 N/m and viscous damping. If the amplitude of free vibration of the mass falls to one half its original value in 2 sec, determine the damping coefficient of the isolator. **[4]**

P.T.O.

Q3) a) Define quality factor and states its significance in frequencies response curve. [8]

b) A single cylinder vertical petrol engine of total mass 400 kg is mounted upon a steel chassis frame and causes a vertical static deflection of 2.5 mm. The reciprocating parts of the engine have a mass of 5 kg and move through a vertical stroke of 120 mm with SHM. A dashpot provided the damping resistance of which is directly proportional to the velocity and amounts to 20 kN at 1 m/s. If a steady state vibrations has been reached. [10]

Determine:

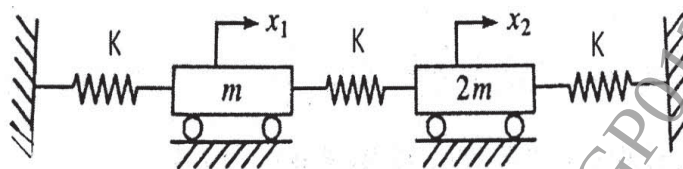
- The amplitude of forced vibrations when the driving shaft of engine rotates at 540 rpm.
- The maximum dynamic force transmitted to the ground through chassis frame (which behaves as a spring). Through the dashpot and through the chassis frame and dashpot together.
- The driving shaft speed at which resonance will occur.

OR

Q4) a) Explain forced vibration with rotating unbalance. [8]

b) The static deflection of an automobile on its springs is 100 mm. Find the critical speed when the automobile is travelling on a road, which can be approximated by a sine wave of amplitude 80 mm and a wavelength of 16 m. Assume the damping to be given by (damping ratio 0.05) also determine the amplitude of vibration at 75 km/hr. [10]

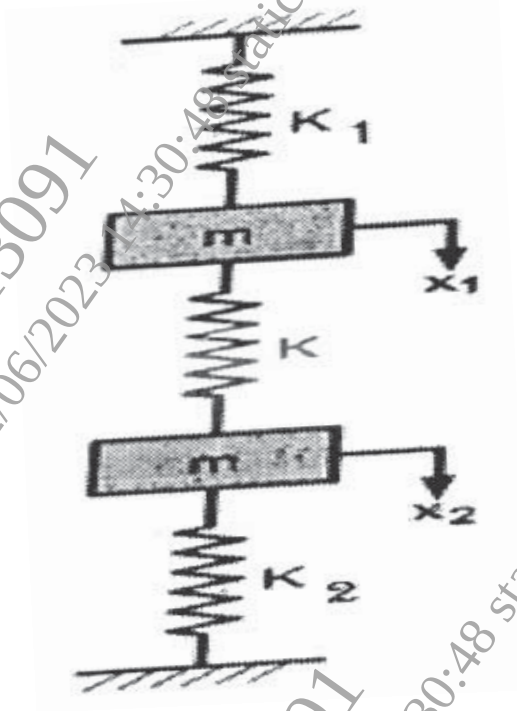
Q5) a) For the system shown in figure, find the natural frequencies of vibration and principal modes of vibration. [10]



b) Derive the concept of torsionally equivalent shaft and derive the equation for its equivalent length. [8]

OR

- Q6) a)** Determine the natural frequencies of the system shown in figure 1 using following data: $K_1 = K_2 = 40 \text{ N/m}$, $K = 60 \text{ N/m}$, $m_1 = m_2 = 10 \text{ kg}$. [10]



- b)** How do you find Eigen value and Eigen vector by Matrix method. [8]

- Q7) a)** What is vibration isolation? Discuss various methods of vibration isolators. [8]
b) Write short notes on: Pass-by-noise. [5]
c) Write short notes on: Noise sources and control of industries. [4]

OR

- Q8) a)** Derive an relation between sound intensity level and sound pressure level. [8]
b) Write short notes on: FFT Spectrum analyzer. [5]
c) Explain with neat sketch the working of sound level. [4]

